

Abraham Barreto

AI & Machine Learning Practitioner

Course: ITAI 2372 — AI Applications

Institution:

San Jacinto College

Focus Areas:

ML, NLP, Computer Vision



Projects & Assignments — Part 1



Machine Learning Model Training

What I Built:

Trained classification & regression models using scikit-learn on real datasets.

Key Skills:

- Data preprocessing & feature engineering
- Model evaluation (accuracy, F1, ROC-AUC)
- Hyperparameter tuning & cross-validation

Tools:

Python · scikit-learn · Pandas · Matplotlib



AI Chatbot / NLP Project

What I Built:

Developed a conversational AI chatbot using NLP techniques and language model APIs.

Key Skills:

- Text tokenization & preprocessing
- Intent recognition & entity extraction
- Prompt engineering & API integration

Tools:

Python · NLTK / spaCy · OpenAI API · Flask

Projects & Assignments — Part 2



Computer Vision / Image Classification

What I Built:

Built a CNN-based image classifier capable of identifying objects from image datasets.

Key Skills:

- Convolutional Neural Network design
- Image augmentation & data pipelines
- Transfer learning with pretrained models

Tools:

Python · TensorFlow/Keras · OpenCV · NumPy



Concepts Mastered

Supervised Learning

Neural Networks & Deep Learning

NLP & Prompt Engineering

Model Evaluation & Optimization

Data Wrangling & EDA

AI Ethics & Responsible AI

How My Skills Translate to Industry Impact

Healthcare

AI-powered diagnostic tools, medical image analysis, and patient outcome prediction using ML models.

Finance

Fraud detection systems, credit risk assessment, and algorithmic trading with classification models.

Retail & E-Commerce

Product recommendation engines, demand forecasting, and customer sentiment analysis via NLP.

Manufacturing

Defect detection on production lines using computer vision and real-time image classification.

Customer Service

Intelligent chatbots and virtual assistants that handle inquiries using NLP and intent recognition.

Technology

Building AI pipelines, automating workflows, and developing smart applications for enterprise use.

LET'S CONNECT

Abraham Barreto

AI Applications · ITAI 2372 Portfolio



GitHub Repository

github.com/AbrahamBarreto/ITAI2372



Email

your.email@example.com



Skills & Certifications

- Python for Data Science & AI
- Machine Learning (scikit-learn)
- Deep Learning (TensorFlow/Keras)
- Natural Language Processing
- Computer Vision (OpenCV)
- ITAI 2372 — AI Applications

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